The Camera, the Split, and the Label: A Three-Check Audit of Glaucoma-Progression Prediction

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Abstract

On the public GRAPE benchmark, deep networks report glaucoma-progression AUCs as high as 0.96. We audit the benchmark's published record and find that, at this sample size, none of the reported numbers can be distinguished from a trivial baseline. We re-evaluate all five published models under one fixed protocol of three checks: a device-only baseline (on GRAPE, the acquisition camera), a patient-grouped split, and a test that the input is disjoint from the window that defines the label. As a positive test, we predict the shipped label from the baseline visit, before its trend is computable. Image features from a frozen fundus model identify the camera at AUC 0.98–1.00, the camera alone predicts progression at 0.56–0.60, and a device-controlled fundus model does no better. Under a fair, device-controlled evaluation every reported number is no better than it, and one model's own released code falls from 0.87/0.96/0.92 to 0.43/0.54/0.55 (PLR2/PLR3/MD). The label is a trend fit to the visual-field series, and we introduce a circularity dial that measures how it is recovered from those defining statistics, with no image, as the input window approaches the window that defines it (AUC 0.90). The checks need only metadata GRAPE already includes: a device label, patient identifiers, the label's definition, and visit dates. We release a fixed split and a reporting template so the audit can be reused.

1 Introduction

Glaucoma is the leading cause of irreversible blindness [31], and its management depends on progression, the question of whether an eye is worsening fast enough to change treatment. The public GRAPE benchmark [14] made this a machine-learning task, pairing fundus photographs and visual-field tests with follow-up. Recent models report strong results: a single baseline color fundus photograph (CFP) reaches AUC 0.71–0.80 [14], models that read a sequence of photographs reach 0.96 (TSDF [29]) and 0.88 (DiffSight-Former [15]), and a model that fuses the visual field with fundus reaches 0.849 [36].

Whether these numbers reflect the eye or the way the data were collected is a question about evaluation, not the model. A single held-out AUC can be high for reasons that have nothing to do with disease. A model can read the acquisition camera instead of the retina [11, 1, 5]. One patient's two eyes can leak across the train and test split [30, 17]. The input can already contain the measurements that define the label. None of these is visible in a single number. The target is contested too: standard visual-field progression algorithms agree on as little as 2.5% of eyes [28], so even a perfect model chases a moving definition.

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We turn these concerns into a compact, reusable audit and apply it to GRAPE and all five published models (Fig. 1). Two kinds of evidence stay separate: the shortcuts, which use no outcome labels and hold on the well-powered endpoints, and the recovered progression signal, which rests on 14 to 40 events [26] and so is read as an interval.

Contributions.

- **The first audit of this benchmark's whole published record** (§3, Fig. 1). Three checks control shortcuts that can produce a high AUC (a device floor, a fair split, and a circularity check), and a forecast from an early visit supplies the positive test. The audit needs only a device label, patient identifiers, and the label's definition, so any progression benchmark can run it. Every one of the five published GRAPE numbers we identified fails at least one check.
- **The device floor** (F1). Image features from a frozen fundus model decode the acquisition camera at AUC 0.98–1.00, the camera alone predicts progression at 0.56–0.60, and a device-controlled fundus model does no better. This shows the documented device-shortcut failure mode sets a ceiling on baseline-fundus progression prediction on GRAPE.
- **Reproduce, then revert** (F2). Fine-tuning the published architectures under their eye-level protocol reproduces the reported range. A fair, device-controlled split leaves them no better than the camera floor, and TSDF's own released code falls from 0.87/0.96/0.92 to 0.43/0.54/0.55.
- **The circularity dial** (F3). The label is a trend fit to the visual-field series, so it can be recovered from those defining statistics with no image (AUC 0.90). Sweeping the input window shows the leakage switch on exactly where the label's definition says it must, when a slope and its p-value first become computable at three visits. Standard circular analysis is diagnosed after the fact. Here the label's definition fixes where the leakage must switch on before we look, and the dial confirms the jump lands exactly there [18].

2 Related work

Glaucoma-progression prediction on GRAPE. GRAPE [14] pairs longitudinal visual fields (VF), fundus photographs, and clinical data for 263 eyes, with progression labels derived from the VF series: PLR2 and PLR3 (pointwise linear regression: progression when at least two or three field locations show a significantly declining slope) and MD (a significantly negative slope of the mean deviation, a global field summary). Its own baseline is a ResNet-50 on a single CFP (AUC 0.71/0.80/0.73). Later work raises the number with fundus sequences (TSDF [29], 0.87/0.96/0.92; and DiffSight-Former [15], 0.79/0.88/0.80) or with a VF+CFP fusion (GPPM [36], per-endpoint AUCs 0.85/0.82/0.69). None controls for the acquisition device, and none reports confidence intervals or checks that the input is disjoint from the label window. Only TSDF releases code, which runs an eye-level `KFold`. Huang leaves the split unstated, and DiffSight, GPPM, and Fouda split at the eye level, where a patient's two eyes can fall on opposite sides of the fold. DiffSight-Former, for instance, evaluates on a single 156/40 eye-level split whose PLR3 test fold holds two progressing eyes. Fouda & Mohammed [10] name "camera artifacts" as a risk without measuring it. We read all five directly and place each against the audit (Table 1, Fig. 1b).

Auditing benchmarks, and leakage. Reproducing published results under a fairer protocol has repeatedly overturned reported progress, in recommendation [9], pruning [2], retrieval [34], and metric learning [22], and meta-reviews catalog how benchmarks mislead through weak baselines, test-set reuse, and leakage [20, 24, 6]. We apply this practice to one clinical benchmark and re-evaluate its entire published record. The closest clinical precedent, a checklist audit of COVID-19 imaging models, found none fit for use [27]. Our three checks turn named leakage categories into concrete tests [17, 4]. A model that correlates the acquisition site or device with the outcome (site-level aggregation leakage) is the device shortcut, shown for radiographs [1, 35, 5], brain MRI [12], and pathology [7] within the wider account of shortcut learning and medical-imaging methodological failure [11, 23, 32]. Badgeley et al. [1] is the closest precedent, recovering the scanner at AUC 1.00 and watching hip-fracture prediction fall to chance once it is controlled. A patient whose two eyes fall on opposite sides of a fold (patient-level aggregation leakage) is the eye-level split, which inflates accuracy across medical imaging [30, 33]. We confirm both on GRAPE and add a measurement of the third.

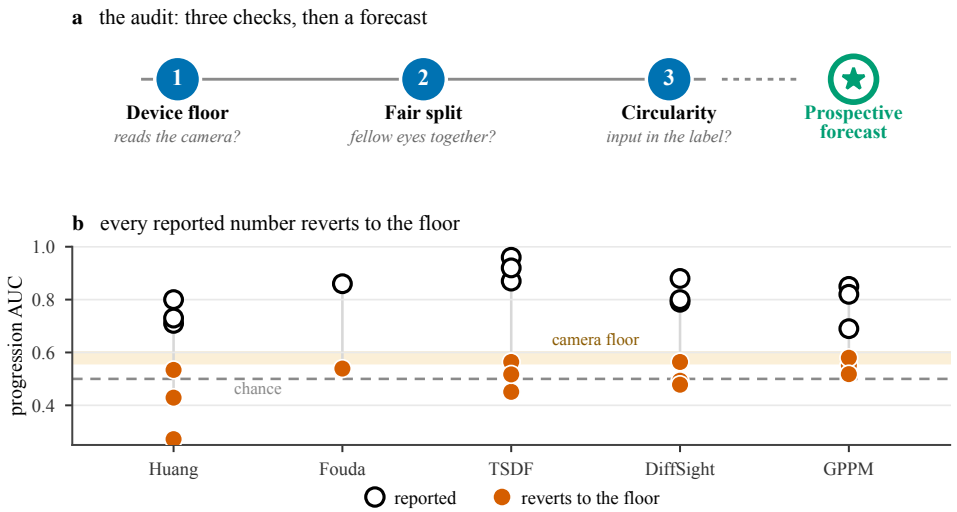


Figure 1: **The three-check audit, and the floor every published number reverts to. (a)** A trustworthy glaucoma-progression claim should clear three checks that catch shortcuts: a device floor (a model given only the acquisition camera), a fair split (fellow eyes kept together), and a circularity check (the input disjoint from the window that defines the label). Beyond them is the prospective forecast we prescribe, the baseline visit predicting the shipped label. **(b)** Every reported number (black, 0.71–0.96) falls to no better than the camera floor (orange, one point per endpoint), where the device, the fellow-eye split, and circularity are controlled together, sitting within the shaded floor band (≈0.56–0.60, wide enough that the sparse 14-event PLR3 can undershoot chance). The per-model reproduce-then-revert behind this teaser is Fig. 3.

Circularity and the contested label. A progression label is a statistic of a time series, and feeding that series to the model invites circularity. Circular analysis, using the same data to select and to analyze, distorts inference whenever the result is not independent of the selection [18]. Ramadan et al. [25] show clinical leakage that appears and disappears as the observation window changes. We make this quantitative for progression, and the dial adds a prediction the observation alone does not: the label's definition fixes the leakage onset in advance (F3). A recent review notes that few glaucoma-progression models share code or test externally [19]. Every primitive we use is prior art [8]. Our contribution is their assembly into one checkable protocol, its first application to a benchmark in current use, and the circularity dial as a new measurement.

3 The audit

Three shortcut-checks and a prospective test. The three checks ask whether a high AUC is riding a shortcut, and a claim passes them only if it clears all three (Fig. 1a). *Check 1 (device floor):* a model given only the acquisition device as input should score at chance. If a device-only model scores well, the device is confounded with the label and any imaging model can exploit it. (On GRAPE the device is the camera, so we call this baseline the *camera floor*.) *Check 2 (fair split):* the reported AUC must survive a fair, patient-grouped, cross-validated evaluation (both eyes of a patient on the same side of every fold, scored across folds rather than on a single favorable split), or the number is not a reliable estimate of the model. *Check 3 (circularity):* the input must be disjoint from the data window that defines the label. Feeding the visual fields whose trend defines progression is predicting the label from itself. Beyond these three we prescribe the positive test a genuine claim should also pass. *The prospective forecast:* predict the shipped label from the baseline (first) visit, before the trend that defines it is computable. This keeps every event, so it is well powered. A stricter variant, the strictly-future landmark, predicts only events that fall after the input window. On GRAPE that leaves too few events to read, so we treat it as a foil, not a main result. Each check uses a field GRAPE already ships: an acquisition-device label, patient identifiers, the label's definition, and visit timestamps, so any benchmark carrying these can run all of them (a device-decode probe, a grouped cross-validation, a window-overlap test, and a baseline-forecast split).

Table 1: **The published GRAPE progression record against the audit.** Reported AUC is PLR2/PLR3/MD. The *Fails* column names the checks the reported number does not clear: a device floor (a model given only the acquisition camera), a fair patient-grouped split, and circularity (the input disjoint from the window that defines the label). Circularity does not apply to a single baseline image (Huang, Fouda) or two early images (DiffSight, below the three-visit slope threshold), which have no serial window to leak. Every reported number fails at least one check, and every reported AUC and split is verified from the five source papers (for TSDF, from its released code). †: weights and key components are unreleased (DiffSight's vessel and optic-disc/cup difference maps, GPPM's evidential fusion), so these two are reproduced as a matched model class. Their verdict is that the eye-level protocol inflates that class, not a re-run of the exact network.

Model	Reported AUC	Fails
Huang et al. [14]	0.71/0.80/0.73	device, split
TSDF [29]	0.87/0.96/0.92	device, split, circularity
DiffSight-Former† [15]	0.79/0.88/0.80	device, split
GPPM† [36]	0.85/0.82/0.69	device, split, circularity
Fouda & Mohammed [10]	0.86 (PLR3)	device, split
Camera-only floor (ours)	0.561/0.600/0.580	*reference*

The checks do not carry equal evidential weight on GRAPE. Check 1's device decode uses no outcome labels, Check 2 is a protocol-level comparison anchored to a training-free floor, and Check 3 reads the continuous visual-field statistics that define the label. Our main conclusions lean on these three and on the well-powered baseline forecast, not on the underpowered landmark.

Metrics. Every AUC we compute is a patient-grouped, repeated five-fold out-of-fold estimate with a patient cluster-bootstrap 95% CI [8]. The eye is the unit, and both eyes of a patient share a fold. The camera-only floor is the same estimator on a single binary device indicator, no pixels. The device-decoding probe is a linear classifier that predicts the acquisition device from a frozen RETFound-CFP [37] embedding, the feature vector the fundus foundation model extracts from each photograph. The device is controlled with a covariate-adjusted AUC [16], which conditions the score on the acquisition device (the binary TOPCON-vs-Canon indicator) so a model cannot use it as a feature. Because the endpoints carry only 14–40 events, we read intervals, not point estimates. These are descriptive comparisons, not confirmatory tests.

Cohort. GRAPE [14] has 263 eyes of 144 patients (all glaucomatous, median 4 visits), with PLR2/PLR3/MD progressors of 40/14/27. Baseline photographs come from three cameras (a TOPCON TRC-NW8 and two Canon CR-2 models), split 157/106 (TOPCON/Canon) by the dataset's own acquisition-device field. The data are public (Figshare collection 6406319). Table 1 places the five published models against the audit.

4 Findings

Each finding makes one claim, with its interval. The four figures trace the audit from shortcut to prospective test.

F1: the acquisition camera is a shortcut, and no representation robustly beats the floor it sets (Check 1). GRAPE's baseline photographs come from a TOPCON TRC-NW8 and Canon CR-2 cameras, and the camera is confounded with the label: TOPCON eyes progress 1.6–$1.9\times$ as often as Canon eyes (Fig. 2a). A model given only a binary TOPCON-vs-Canon indicator (no image content) therefore predicts progression at AUC $0.561/0.600/0.580$ (Fig. 2), a training-free floor that uses no pixels. A frozen RETFound-CFP embedding distinguishes TOPCON from Canon at out-of-fold AUC 0.98, rising to 1.00 on the optic-disc crop the downstream papers actually use. The device is therefore trivially available to any model, and cropping does not remove it. The naive fundus-embedding AUC (0.57–0.60) is no higher than the camera floor, and controlling the device (a covariate-adjusted AUC) leaves it at the floor (Appendix B). No published model controls for the device, though the acquisition field ships with the dataset. The camera does not photograph sicker

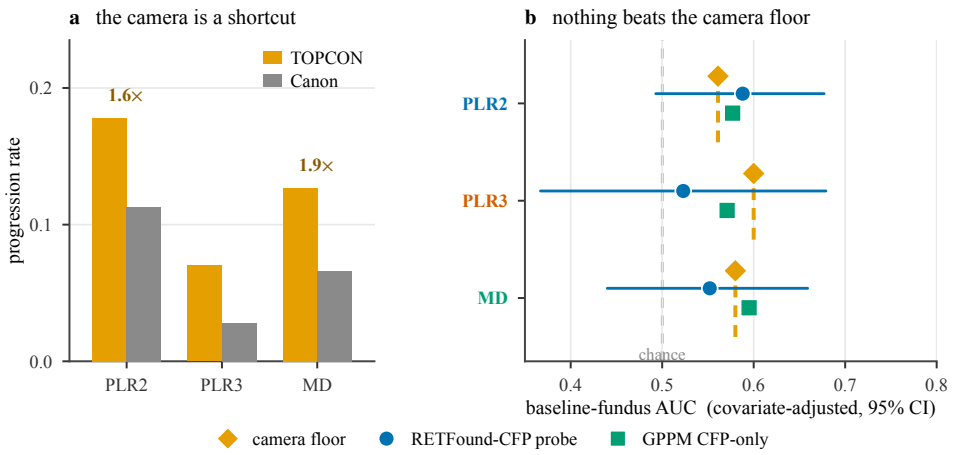


Figure 2: **The device confound and the floor it sets (Check 1, F1). (a)** The acquisition camera is a usable shortcut: TOPCON eyes progress 1.6–1.9× as often as Canon (and more often on the sparse 14-event PLR3 endpoint too), a frozen embedding decodes the camera at out-of-fold AUC 0.98–1.00, and the camera alone (no pixels) predicts progression at 0.56–0.60. **(b)** Every device-controlled estimate of the baseline-fundus signal then coincides with that floor (orange): a covariate-adjusted RETFound-CFP probe and GPPM's CFP-only baseline. On every endpoint (PLR2/PLR3/MD) no 95% CI clears the floor.

eyes: baseline severity (visual field, RNFL, IOP) is matched across cameras, and the confound instead tracks follow-up length (TOPCON eyes are followed longer, so their slope-defined labels trigger more often), an exposure-time artifact rather than retinal prognosis (App. B).

The device-controlled baseline-fundus signal remains near the floor. A frozen RETFound-CFP disc-crop probe (the crop the downstream papers use), covariate-adjusted for the device, scores 0.59/0.52/0.55, the CFP-only baseline of an independent group and architecture (GPPM's, a generous eye-level upper bound) sits at the floor (App. A), and the fine-tuned single-CFP models the GRAPE and Fouda papers train are likewise at the floor once the device is controlled (Fig. 3). A sweep over eight representation-and-crop combinations agrees: every one's interval includes or sits below the floor, and the one exception (a ConvNeXt-Base probe on MD, 0.74 [0.66, 0.82]) does not replicate across backbone scale (ConvNeXt-T/B/L give 0.49/0.74/0.61 on MD) or under fine-tuning (Table 3). No representation we tried clears the floor robustly, so the embedding that identifies the camera at AUC 1.00 carries no baseline-fundus signal we can distinguish from the device. The floor itself carries a wide interval that spans chance (App. A), so at these event counts these reads are non-inferiority to the floor, not proof that no fundus signal exists.

F2: the reported range reproduces eye-level, then reverts to the floor (Check 2). We fine-tuned the single-CFP architectures the GRAPE and Fouda papers use, with their imbalance recipe (App. A). Under the standard eye-level split these papers use or imply, the best of our seed runs reaches the published numbers, but those values sit at the top of a wide seed spread whose medians are only 0.41–0.70 (Fig. 3): the reported number is the high end, not the center. Under a fair split the same models revert to 0.43–0.56, no better than the camera floor. PLR3 is especially fragile: with 14 progressors an eye-level fold holds about three positives, so its AUC takes only a few discrete values, and both the published 0.80 and Fouda's 0.86 sit within that seed-to-seed noise. The reversion is not an under-training artifact: the same recipe reaches the published range eye-level while a frozen probe and a fine-tuned single-CFP model both sit at the training-free floor (F1), so the drop is the split, not the budget.

The serial-image models reproduce, then revert, on their own code. TSDF's released code [29] runs an eye-level `KFold` and selects the best epoch on the test fold, a selection-bias inflation of the estimate [3]. On its own released features we recover its numbers to within 0.002 (0.866/0.954/0.917), and fixing three defects (a fair split, held-out epoch selection, and a pooled out-of-fold AUC) reverts them to 0.43/0.54/0.55 at the camera floor (Fig. 3). This is a property of the protocol and of the

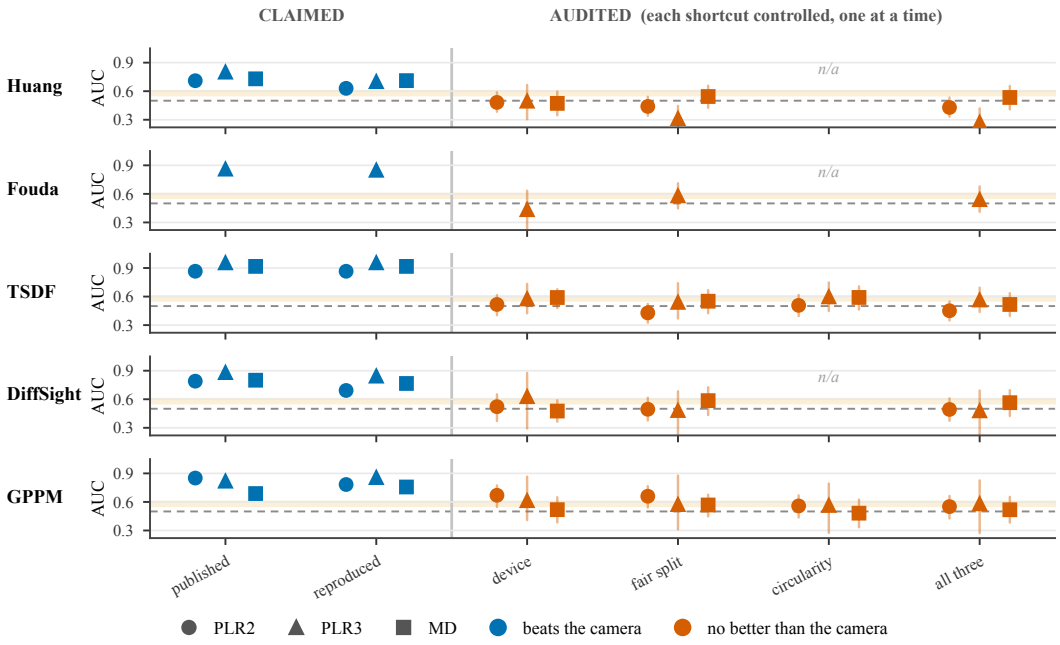


Figure 3: **Reproduce, then revert: every published number is no better than the camera once each shortcut is controlled.** One row per method, reproduced by its own released code or architecture where available (ResNet-50 for Huang, a weighted CNN ensemble for Fouda, the released code for TSDF) and by a matched model class where weights are unreleased (a two-image change head for DiffSight, GPPM's two-field VF+CFP mechanism). For the latter two, the result is that the protocol inflates the model class, not that their exact network lacks signal. Each point is an endpoint (circle PLR2, triangle PLR3, square MD; PLR3 has only 14 events, about three per fold, so its single values are noisy). A marker is blue when its 95% interval clears that endpoint's own camera floor (*beats the camera*) and orange when the interval includes it (*no better than the camera*). Whiskers are intervals. The CLAIMED columns give the published AUC and our reproduction, which recovers it. The AUDITED columns control one shortcut at a time, the device (a covariate-adjusted AUC [16]), a fair patient-grouped split, and a non-circular input, and then all three together. The four image models are no better than the camera under every control. GPPM comes closest: its device- and split-controlled points on PLR2 are the highest of the five (about 0.67), but no interval clears the floor, and it reaches the floor once the input is made non-circular. *n/a*: the check does not apply, because a single- or two-image input has no serial window to leak.

serial-image class, not one model: a sequence of serial-image variants, from change features to a temporal attention head, reproduces the reported range eye-level and sits at the patient-grouped floor at every step (App. C). DiffSight-Former [15] reports 0.79/0.88/0.80 on a single 156/40 eye-level split with two PLR3 test positives. Reproducing its two-image input in its own regime, we reach those numbers at the eye-level upper tail and drop to the floor (0.49/0.48/0.59) under a fair split (App. C). Every serial number thus reflects the split rather than the forecast its design intends.

F3: the circularity dial (Check 3). The progression labels are a trend fit to an eye's visual-field series, so feeding that series to predict the label is predicting the label from itself. Circular analysis warns that using the same data to define and to estimate a quantity distorts it whenever the two are not independent [18]. Here the quantity is the label and the data are the visual fields. We turn that warning into a measurement. A *circularity dial* tracks the leakage as the input window lengthens toward the window that defines the label (Fig. 4a): predicting the fixed MD label from the visual-field statistics of the first k visits climbs from 0.47 at one visit to 0.90 once the whole series is available, and the jump lands exactly at k=3, when a slope and its p-value first become computable. That threshold is sharp: a slope and its p-value cannot be computed from fewer than three visits (two points give a slope but no significance), so an input spanning one or two visits cannot recover the label-defining statistic and stays near chance. The onset is a property of the label,

not of counting visits: loosening the slope's significance threshold ($p < 0.10$) leaves the jump at k=3, while a placebo label defined by a two-visit change (computable at k=2) has its jump at k=2 instead (App. D). A two-field pair lifts the score by a separate, weaker route. The pair recovers the label only when the second field falls inside the label-defining window, so it is the time span that leaks, not the slope (Fig. 4b). A statistic ablation confirms the mechanism: the label-defining slope (AUC 0.91) and its p-value (0.95) carry the 0.90, while recency, variability, and visit count do not (Fig. 4c). The dial is general: any benchmark whose label is a trend fit to a series can be probed the same way, providing a direct test for a familiar pattern. Progression AUCs often rise with the number of input visits, and this is usually read as a model accumulating evidence. Read this way, the rise is instead leakage accruing as the input window grows into the label-defining window. It affects any model whose input reaches into that window, and most directly a visual-field-history model, which reads the defining statistics themselves.

GPPM's fusion is the clearest instance, and we reproduce its mechanism on GRAPE. It pairs a baseline visual-field test with one follow-up test (its fundus branch is down-weighted to 0.01, so the visual field carries the number), and a two-field pair recovers the fixed label only when the second test is late: the baseline-and-last pair scores 0.83/0.86/0.86, while a baseline-and-second pair (both fields early, before the label window) sits at chance (0.51–0.53). Sweeping the second test later lifts the AUC with its time span, not the count of two (Fig. 4b). This is about what GPPM's input carries, not its architecture: its own fundus-only branch already sits at the camera floor (App. A), so the fusion's lift is in-window field change, not baseline prognosis, and the same early pair forecasts a strictly later event no better than chance. The serial-fundus models are a weaker, less clear-cut case. TSDF consumes the full fundus visit sequence, which reaches into the same window, so its lift is at least partly structural change occurring during that window rather than a prospective forecast (App. C). DiffSight-Former frames its task as forecasting future onset, a claim GRAPE's short series cannot adjudicate.

The prospective forecast we prescribe stays at the floor. The positive test that no paper runs is to predict the shipped label from the baseline visit, before the trend that defines it is computable. This forecast is non-circular and keeps the full event counts (40/14/27), so it is well powered on the PLR2 and MD arms. On GRAPE it cannot beat the camera floor: the baseline fundus sits at the floor (0.60/0.57/0.59), and a multimodal fusion of the visual field, optical coherence tomography (OCT), fundus, and intraocular pressure (IOP) is no better (App. C). The baseline forecast is thus at the floor even as the circular in-window number reaches 0.90, so GRAPE places progression at the floor while it stays forecastable at scale (§5).

5 Discussion and limitations

What this means for the field. We read these as failures of the evaluation setup, not of the teams behind these models. The confound comes from scoring a device-confounded, visual-field-labeled public benchmark by a single leaderboard AUC, a pitfall the shortcut-learning literature documents across medical imaging [11, 35]. To the resolution the data allow, the published numbers are protocol artifacts: a device shortcut, an eye-level split, and label circularity account for the reported range where we reproduce it (Fig. 3), and no estimate we tried clears the floor. This does not say progression is unpredictable: in large, device-controlled cohorts baseline models forecast trajectory progression at AUC 0.84 (0.75 on an external device) [21] and glaucoma *onset* at ≈0.74 [13]. But GRAPE, at 14–40 events with a device-confounded, VF-derived label, cannot establish it: its numbers measure the protocol, not the eye.

What to do instead. The audit suggests one practical fix for each check. For the device, report the camera floor and a covariate-adjusted AUC. For a fair split, keep each patient's two eyes on the same side of a cross-validated split. For circularity, keep the model input disjoint from the label-defining window. And report a baseline prospective forecast, the first visit predicting the label, alongside any in-window number, each with a cluster-bootstrap interval at the stated event count. These changes make a leaderboard number easier to interpret as a clinical claim. We release a fixed patient-grouped split and a reporting template so the protocol can be applied consistently (App. F).

Limitations. *Small event counts.* With 14–40 events the per-endpoint intervals are wide. We lean on the event-independent device decode, the training-free floor and split comparison, and the well-

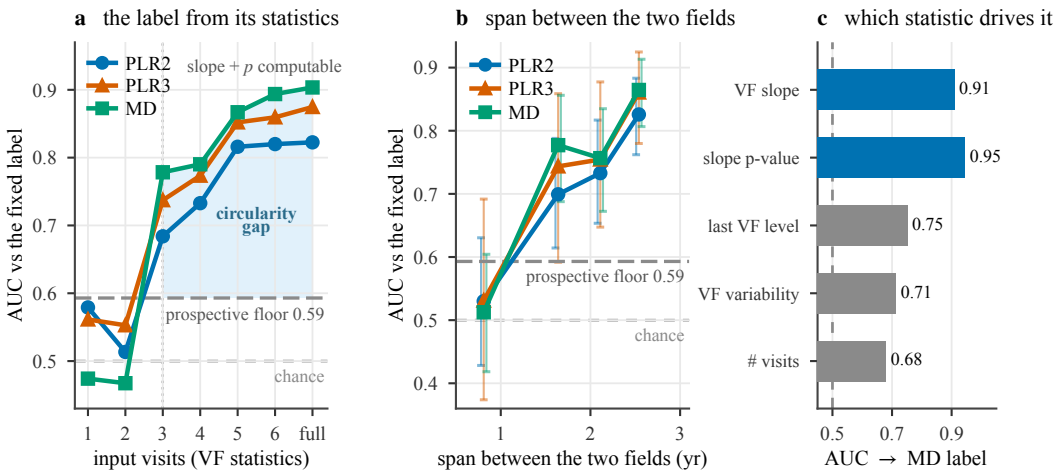


Figure 4: **The 0.90 comes from predicting the label from its own statistics (Check 3). (a)** Out-of-fold AUC for predicting the fixed label GRAPE provides from the visual-field statistics of the first k visits, as k grows into the window that defines the label. The MD AUC climbs $0.47 \rightarrow 0.90$, jumping once the slope and its p-value become computable ($k{=}3$). The baseline prospective forecast (first visit $\rightarrow$ the shipped label, dashed) sits at the camera floor (≈ 0.59, on the full event counts 40/14/27), and the shaded band between it and the circular curves is the circularity gap. **(b)** The two-field model (GPPM's mechanism): a pair of visual-field tests recovers the fixed label only when the second test is late. As the span between the two tests grows (baseline+2nd $\rightarrow$ baseline+last), the AUC climbs from chance to 0.86 (95% CI whiskers), so it is the span that leaks, not the count of two. The early pre-window pair sits at chance, and the baseline prospective forecast (dashed) stays at the floor. **(c)** Full-series AUC from each statistic alone: the label-defining slope (0.91) and its p-value (0.95) carry the 0.90, not recency, variability, or visit count.

powered PLR2 and MD arms, and corroborate the floor with several estimates rather than one. *A single cohort.* GRAPE is one all-glaucoma dataset. The audit is cohort-agnostic, an independent group's CFP-only baseline already reproduces our floor [36], and the mechanisms it exposes (device shortcut, fellow-eye leakage, label circularity) are established in the ML literature [11, 17], so running the audit on a second multi-site cohort is the natural validation and future work. *VF-derived labels.* Every GRAPE label is a visual-field trend. That circularity is one of our findings (F3) but also precludes a genuine structural endpoint here. *Serial reproduction.* We reproduce TSDF on its released code and DiffSight as a matched model class (its weights are unreleased) on GRAPE's short image sequence. Both revert to the floor (App. C).

6 Conclusion

A glaucoma-progression AUC on GRAPE can be high because a model reads the camera, because a patient's two eyes leak across the split, or because the input already contains the label, and none of these is visible to a single-split AUC. Our audit makes these sources visible with three checks, a device floor, a fair split, and a circularity dial that reads the label leaking back into the input as the window grows, and adds a baseline prospective forecast for the prediction that would actually count. Applied to GRAPE, every published progression number fails at least one check, and the baseline-fundus signal cannot, at this sample size, be distinguished from a training-free camera floor (0.56–0.60 across endpoints). On GRAPE today, a high score can still come from the camera, the split, or the label. Until a number clears all three checks and also forecasts a held-out future, it is not yet evidence about the eye.

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A Audit implementation and cohort details

Estimator and intervals. Every AUC we compute is a patient-grouped, repeated five-fold out-of-fold estimate from an ℓ_2-regularized logistic model, with a patient cluster-bootstrap 95% CI (patients, not eyes, are the resampled unit) [8]. Eyes are scored, but both eyes of a patient always share a fold, so no eye of the same patient straddles the split. Tabular modalities are the pre-specified summaries GRAPE ships: the visual field, the retinal-nerve-fiber-layer (RNFL) thickness from optical coherence tomography (OCT), and intraocular pressure (IOP). The fundus embedding is a frozen RETFound-CFP [37] representation reduced to 16 principal components. The device is controlled with a covariate-adjusted AUC [16], the standard estimator for holding a covariate fixed when comparing discrimination: it conditions the score on the acquisition device (binary TOPCON-vs-Canon), so a model cannot use the device as a feature.

The camera-only floor and the device probe. The camera-only floor is the same logistic estimator trained on a single binary device indicator (is-TOPCON) and no image content; it is training-free in the sense that it uses no pixels and no learned representation. The device-decoding probe scores the frozen fundus embedding out-of-fold against the device label; we report it on the full-fundus crop (AUC 0.98) and on the optic-disc crop the downstream papers use (1.00). We use three fundus crops throughout: the full field of view ("full"), a crop centered on the optic disc ("disc", the crop that decodes at 1.00 and the one the fundus probe uses), and a tighter disc region-of-interest ("ROI"); Table 3 reports all three. The camera-only floor carries its own patient cluster-bootstrap CI (0.56 [0.48, 0.64] / 0.60 [0.48, 0.71] / 0.58 [0.49, 0.66]), and every device-controlled fundus read lies within it (§4).

The GPPM CFP-only number. GPPM [36] reports two fundus-only disc region-of-interest (ROI) reads in its Table 3: an independent evidential-deep-learning *Baseline* (a separate architecture GPPM adopts, not its own branch) at $0.577/0.571/0.595$, the value we use in the floor triangulation (F1), which sits at the camera floor on all three endpoints. The second is GPPM's own class-balanced CFP branch (*CFP_M*) at $0.668/0.727/0.555$, whose higher PLR2/PLR3 are biased artifacts (zero sensitivity on PLR3 and MD, a biased AUC on 6–16 positives). Its headline 0.849 is a visual-field+CFP fusion under a combined criterion, not a CFP-only result, so the CFP-only read we cite is the reliable one and it sits at the camera floor. This number is reported under GPPM's own eye-level, device-uncontrolled, 196-eye protocol. Since an eye-level split and an uncontrolled device only inflate an AUC, 0.58 is a generous upper bound on GPPM's baseline-fundus signal (a patient-grouped, device-controlled evaluation could only lower it), so its coincidence with our camera floor is a conservative comparison, not a like-for-like one.

Reproducing GPPM's two-field mechanism. GPPM pairs a baseline visual-field test with a single follow-up test and a fundus photograph (the paper leaves the fundus visit timing under-specified), down-weighting the fundus branch to 0.01, so the visual-field pair carries its number. We reproduce that mechanism directly on GRAPE's visual-field series; no fusion architecture is needed, since the field decline is the signal. From five features of a two-field pair (the two mean-sensitivity levels, their difference, the two-point rate, and the span between them) we predict the fixed PLR2/PLR3/MD label, patient-grouped. Pairing the baseline with the last visit (GPPM's in-window structure) scores $0.83/0.86/0.86$. Pairing it with the second visit, the same two tests but both early, sits at chance ($0.53/0.53/0.51$). The intermediate pairs rise monotonically as the second test moves later (mean span $0.8\rightarrow2.5$ years), so the lift tracks how far the second test reaches into the label window, not the count of two. The same early pair predicting a transparent MD-slope event on the disjoint later window (visits 3–end) cannot be distinguished from chance (0.31 [0.00, 0.82], 5 held-out events), consistent with the single-field landmark (a fixed-time forecast). This reproduces GPPM's two-field mechanism, not its evidential-fusion architecture. Its own CFP-only branch already sits at the floor (above), so the fusion adds in-window functional change, not baseline prognosis (`grape_two_vft.json`). The visual-field labels also inherit the large test–retest variability of perimetry, which worsens with field damage. This adds noise to the trend fit and caps the AUC achievable by any prognostic model, over and above the small event counts.

Reproduction recipe (F2). We fine-tuned ResNet-50, EfficientNet-B4, DenseNet-121, and ConvNeXt-Tiny, plus a soft-vote ensemble, with the Fouda-style imbalance recipe (optic-disc ROI

crop, focal loss α=0.75, γ=2.0, class-balanced sampling, last-30% fine-tune, 120 balanced mini-batches of 32 [15 epochs $\times$ 8 batches]). The *published* protocol is a single random eye-level 80/20 split over five seeds (no patient grouping, no device control, no cross-validation); the *patient-grouped* protocol is a five-fold pooled out-of-fold (OOF) estimate with the camera-only reference. Eye-level entries in Table 2 are the median (and max) over seeds. The reversion is not an under-training artifact: the *same* recipe reaches the published range on the eye-level split, the camera floor is training-free, and a frozen linear probe and a fine-tuned single-CFP model both sit at the same floor (F1), so the drop is the split, not the training budget.

Compute. All audit analyses (frozen embeddings, logistic probes, window-length sweep, landmark, bootstrap) run on CPU in minutes; the F2 fine-tuning runs on a single consumer GPU. No result requires re-scoring to reproduce the figures: every number is read from a committed data panel.

B The device confound in detail

The camera is confounded with the label (Fig. 2a), so a device-only model scores $0.561/0.600/0.580$ with no pixels. The naive disc-crop fundus-embedding AUC is no higher than that floor, and controlling the device leaves it at the floor: PLR3 drops from 0.57 to 0.52 and MD from 0.59 to 0.55, with a weak PLR2 residual (0.59, its interval including the floor). The full representation ladder (Table 3) shows the same across eight representations and three crops. Because the acquisition-device field ships with GRAPE, this confound is controllable, but no published model controls for it. Is the camera indexing genuinely more-advanced eyes (in which case a fundus model reading severity would be doing real prognosis), or is it an exposure-time artifact? We checked directly (`grape_camera_casemix.json`): baseline severity is matched across cameras, TOPCON vs Canon median baseline VF mean sensitivity 22.98 vs 22.26, RNFL 80 vs 81, IOP 16.5 vs 18.5 (all Mann–Whitney $p > 0.05$), so the camera does not photograph sicker eyes. The one metric that approaches a difference, IOP (p=0.068), runs higher in the *less*-progressing camera (Canon), so it argues against a case-mix explanation rather than for one. But TOPCON eyes are followed markedly longer (median 3.12 vs 2.03 years, $p < 0.001$), and a slope's significance is mechanically easier to reach with longer follow-up. The camera-progression association is thus an exposure-time artifact, not baseline case-mix and not retinal prognosis, which sharpens rather than weakens the point.

C Serial fundus does not forecast

The serial-image models that report 0.87–0.96 feed a fundus *sequence*. On GRAPE's follow-up photographs, even the in-window serial-fundus signal (whose input overlaps the label window) is at or below chance on all three endpoints, naive or device-controlled, and the follow-up embedding identifies the acquisition device at AUC 0.96 while 41% of eyes switch camera between visits. A serial-fundus model on GRAPE is therefore reading a changing device at least as much as a changing eye, and there is no prospective fundus forecast here. The published numbers come from the evaluation protocol, which we reproduce directly below.

Direct reproduction of TSDF. TSDF's released code (`github.com/zongzi13545329/MICCAI`) provides a direct check of the mechanism. It trains a DSMIL (dual-stream multiple-instance learning) aggregator on the full per-eye visit sequence of precomputed MAE (masked-autoencoder) features (in-window by construction: every follow-up visit is fed, so the input spans the label window) under an eye-level `KFold(shuffle, random_state=42)` with unweighted cross-entropy. It then reports the mean over folds of each fold's best-epoch test AUC, with the best epoch chosen on the test fold itself, so epoch selection is not held out. Re-running that exact recipe on its own released features recovers $0.866/0.954/0.917$ (vs. published $0.866/0.956/0.917$; one PLR3 fold reaches 1.00 on two positives). Fixing the three defects independently (a `GroupKFold` by the 144 patients, epoch selection on a patient-disjoint validation split, and a pooled out-of-fold AUC with a patient cluster-bootstrap CI) reverts them to 0.43 [0.33, 0.52] / 0.54 [0.37, 0.74] / 0.55 [0.42, 0.67] (this is the split-controlled read; controlling the device and circularity together as well leaves it at $0.45/0.56/0.52$, Table 4). A ladder of fundus-only serial models (a first-two-image structural-difference model $\rightarrow$ simple change features $\rightarrow$ pooled RETFound embeddings $\rightarrow$ a temporal attention head) reproduces the range as the eye-level max over seeds and sits at the patient-grouped floor at every rung, so

the reversion is a property of the protocol, not the model. This rung is a model-class stand-in for DiffSight-Former, not its architecture: on GRAPE DiffSight takes the first two fundus images per eye (its $\leq$5-visit pre-onset prospective forecasting is its separate experiment on the SIGF onset dataset, not GRAPE) and encodes inter-visit change through vessel and optic-disc/cup difference maps, none of which is released. With only two visits the input is below the three-visit slope threshold, so it is non-circular but not prospective. We reproduce the part that can be tested from the available materials: a small learned attention head over the *standardized* frozen RETFound features of those two images, letting the head model the inter-visit change rather than injecting a hand-made difference, with *no* hand-engineered features. It is trained in DiffSight's own regime (mini-batch, cosine annealing with warm restarts, plain cross-entropy). Train-fold feature standardization conditions the head and lifts the whole seed distribution off its earlier below-chance floor. Between the two protocols we change a single knob, the split, and hold the model, input, and training fixed. Swept over eye-level seeds this head reaches all three reported numbers at the upper tail of the seed distribution (PLR2/PLR3/MD max $0.82/0.97/0.90$ vs. its $0.79/0.88/0.80$, on $5/2/3$ test positives; the reported values sit in the top decile of the seed distribution), reproducing DiffSight's single-split numbers as the tail of a leaky regime. Changing only the split to patient-grouped, the same head drops to the floor: 0.49 [0.38, 0.62] / 0.48 [0.23, 0.68] / 0.59 [0.43, 0.73], every interval spanning chance and none clearing the camera floor. Because the reproduce and the revert differ in nothing but the split, the reported number is a property of the split, not the image content. DiffSight's weights and its vessel/disc segmentation maps are unreleased, so this measures its regime and model class rather than its exact architecture.

The baseline prospective forecast (well-powered). The prescribed positive test predicts the shipped label from the first visit, before the label's trend is computable, so it keeps the full event counts ($40/14/27$) and is well-powered, unlike a strictly-future landmark. We run it two ways, patient-grouped and device-controlled. *Image-only* (fundus alone): the first visit is the single-CFP disc probe ($0.60/0.57/0.59$), the first two images the DiffSight two-visit head ($0.49/0.48/0.59$); both sit at the floor. *Fusion* (visual field, OCT-RNFL, fundus morphometry, and IOP): the first visit scores $0.58/0.56/0.59$, every 95% interval spanning chance, and extending the input to the first two visits stays at the floor. This also completes the serial-model audit non-circularly: restricting each serial model to its first $\leq$2 visits is the image-only read (TSDF, DiffSight) or the fusion read (GPPM's baseline+early visual-field+fundus), all at the floor. A fundus-specific TSDF cannot be run more prospectively than this on GRAPE: its fundus follow-up is too sparse ($\approx$1 pre-window image per eye), so TSDF's high number intrinsically requires the in-window sequence. The first-visit read is cleanly non-circular; the first-two read adds an early-trajectory point that stays below the three-visit significance threshold.

D Dial robustness: the onset tracks the label, not the visit count

If the dial's jump were an artifact of "more visits, more signal", it would sit at the same k for every label. It does not: it tracks when the label's defining statistic first becomes computable. Recomputing the dial with the same window-feature, patient-grouped out-of-fold pipeline for alternative labels, the onset moves with the definition. On the MD-slope label the jump is at k=3 under the shipped $p < 0.05$: the AUC rises from 0.47 [0.38, 0.55] at k=2 to 0.78 [0.67, 0.88] at k=3, disjoint intervals, and the jump stays at k=3 under a looser $p < 0.10$ (0.66 [0.57, 0.74] to 0.86 [0.80, 0.93]). On a placebo label defined by a two-visit change (computable at k=2) the jump moves to k=2: 0.38 [0.28, 0.49] at k=1 to 1.00 [0.99, 1.00] at k=2 (the window slope at k=2 is exactly the placebo's defining statistic, hence near-perfect recovery). The intervals are disjoint across each onset, so it is predicted a priori by the label, which is the sense in which the dial is a diagnostic rather than an observed curve (`grape_dial_robustness.json`).

E Full result tables

These tables give the exact numbers behind the figures. Table 2 (reproduce-then-revert, single-CFP architectures) backs Fig. 3, and Table 3 (the covariate-adjusted representation ladder) backs the floor-as-ceiling claim (F1). Table 4 gives the per-model audited AUCs with intervals behind Fig. 3, and Table 5 gives the circularity dial and per-statistic values behind Fig. 4.

Table 2: **Reproduce-then-revert (Check 2).** Fine-tuning the architectures the GRAPE/Fouda papers use, with their imbalance recipe, under the published eye-level protocol reaches the published range (the seed distribution's upper tail brackets the single published numbers), then reverts to the training-free camera-only floor under a patient-grouped split. Eye-level entries are median (max) over 5 seeds; grouped entries are patient-grouped five-fold pooled OOF. All values are AUC.

	PLR2 (40 ev)		PLR3 (14 ev)		MD (27 ev)	
Model	eye-level	grouped	eye-level	grouped	eye-level	grouped
ResNet-50	0.41 (0.71)	0.45	0.57 (0.72)	0.48	0.51 (0.57)	0.52
EfficientNet-B4	0.57 (0.66)	0.47	0.61 (0.83)	0.44	0.45 (0.62)	0.43
DenseNet-121	0.53 (0.72)	0.51	0.68 (0.83)	0.56	0.70 (0.80)	0.51
ConvNeXt-Tiny	0.43 (0.51)	0.52	0.55 (0.71)	0.46	0.41 (0.50)	0.50
Soft-vote ens.	0.52 (0.69)	0.47	0.67 (0.77)	0.44	0.50 (0.53)	0.49
published (lit.)	0.71		0.80 · 0.86 ens		0.73	
camera-only floor	0.56		0.60		0.58	

Table 3: **Representation ladder (Check 1).** Covariate-adjusted, patient-grouped repeated-OOF AUC [95% CI] of frozen linear probes over eight representation/crop combinations, beside the training-free camera-only floor. Every representation's interval includes or sits below the floor on all three endpoints, except the lone ConvNeXt-B MD point whose lower bound clears it; that crossing does not replicate on ConvNeXt-L or under fine-tuning (§4).

Representation (frozen LP, covariate-adjusted)	PLR2	PLR3	MD
RETFound-CFP (full)	0.504 [0.39, 0.60]	0.584 [0.40, 0.74]	0.558 [0.44, 0.67]
RETFound-CFP (disc)	0.588 [0.49, 0.68]	0.523 [0.37, 0.68]	0.552 [0.44, 0.66]
RETFound-CFP (ROI)	0.550 [0.45, 0.65]	0.646 [0.51, 0.78]	0.470 [0.36, 0.59]
DINOv2-reg (full)	0.286 [0.20, 0.37]	0.593 [0.44, 0.74]	0.480 [0.35, 0.60]
DINOv2-reg (disc)	0.480 [0.39, 0.58]	0.458 [0.30, 0.62]	0.556 [0.41, 0.69]
ConvNeXt-T (disc)	0.564 [0.46, 0.66]	0.626 [0.47, 0.76]	0.489 [0.38, 0.60]
ConvNeXt-B (disc)	0.593 [0.51, 0.67]	0.649 [0.46, 0.81]	0.744 [0.66, 0.82]
ConvNeXt-L (disc)	0.522 [0.43, 0.60]	0.523 [0.40, 0.64]	0.610 [0.51, 0.71]
camera-only floor	0.561	0.600	0.580

F A standardized re-evaluation protocol

So that the audit is adoptable rather than only diagnostic, we release both a fixed evaluation split and a reporting template. *The split.* `grape_standard_splits.csv` assigns each of the 263 eyes to one of five folds by a deterministic patient-grouped `GroupKFold` (144 patients; both eyes of a patient in the same fold, so fellow-eye leakage is impossible; one split serves PLR2/PLR3/MD). Scoring on these fixed folds removes the single-favorable-split degree of freedom. *The template.* On these folds, a trustworthy progression claim reports: (i) the training-free camera-only floor beside the model, and a device-adjusted AUC (we use a covariate-adjusted AUC); (ii) a patient-grouped, pooled out-of-fold AUC rather than a single split; (iii) a baseline prospective forecast (the first visit predicting the shipped label) alongside any in-window number; (iv) a patient cluster-bootstrap 95% CI and the event count for every endpoint; and (v) whether the model input overlaps the label-defining window. Each item is a one-line addition, and together they turn a leaderboard number into a defensible clinical claim.

G Reproducibility and broader impact

Reproducibility. GRAPE is public under CC BY 4.0 (Figshare collection 6406319), and the RETFound-CFP weights we probe are used non-commercially under CC BY-NC 4.0. On acceptance we release the analysis panel and the two regeneration scripts under the MIT license. Every number in the paper is read from a single committed data panel with a reconciliation gate, and the figures and tables are regenerated from it by two scripts with no model calls or re-scoring; the audit estimators (device floor, device-decode probe, circularity sweep, baseline prospective forecast,

Table 4: **Per-model audited AUCs [95% CI] behind Fig. 3.** Each reported number, reproduced by its own architecture, once each shortcut is controlled one at a time: the device (a covariate-adjusted AUC), a patient-grouped split, a non-circular input, and all three together. *n/a*: the control does not apply, because a single- or two-image input has no serial window to leak; Fouda reports PLR3 only. Compare against the camera floor ($0.561/0.600/0.580$): no audited read clears it. GPPM's device and split reads are the highest of the five, but their intervals include the floor.

Model	Control	PLR2	PLR3	MD
Huang	device	0.48 [0.38, 0.59]	0.49 [0.30, 0.67]	0.47 [0.35, 0.60]
	fair split	0.44 [0.34, 0.54]	0.31 [0.19, 0.45]	0.54 [0.42, 0.66]
	circularity	*n/a*	*n/a*	*n/a*
	all three	0.43 [0.33, 0.53]	0.27 [0.15, 0.42]	0.53 [0.41, 0.66]
Fouda	device	–	0.43 [0.23, 0.63]	–
	fair split	–	0.58 [0.45, 0.71]	–
	circularity	–	*n/a*	–
	all three	–	0.54 [0.41, 0.68]	–
TSDF	device	0.52 [0.40, 0.62]	0.57 [0.42, 0.73]	0.59 [0.48, 0.68]
	fair split	0.43 [0.33, 0.52]	0.54 [0.37, 0.74]	0.55 [0.42, 0.67]
	circularity	0.51 [0.39, 0.62]	0.60 [0.45, 0.75]	0.59 [0.46, 0.71]
	all three	0.45 [0.35, 0.55]	0.56 [0.44, 0.69]	0.52 [0.40, 0.64]
DiffSight	device	0.52 [0.37, 0.65]	0.63 [0.29, 0.88]	0.48 [0.36, 0.59]
	fair split	0.49 [0.38, 0.62]	0.48 [0.23, 0.68]	0.59 [0.43, 0.73]
	circularity	*n/a*	*n/a*	*n/a*
	all three	0.49 [0.37, 0.61]	0.48 [0.22, 0.69]	0.56 [0.42, 0.70]
GPPM	device	0.67 [0.55, 0.77]	0.61 [0.41, 0.87]	0.52 [0.38, 0.65]
	fair split	0.66 [0.54, 0.77]	0.57 [0.31, 0.88]	0.57 [0.45, 0.68]
	circularity	0.56 [0.44, 0.67]	0.56 [0.28, 0.79]	0.48 [0.33, 0.62]
	all three	0.55 [0.43, 0.66]	0.58 [0.27, 0.83]	0.52 [0.38, 0.65]

Table 5: **The circularity dial behind Fig. 4.** *Top:* out-of-fold AUC for predicting the fixed label from the visual-field statistics of the first k visits, climbing to 0.90 with the jump at $k{=}3$, when a slope and its p-value first become computable. *Bottom:* full-series AUC from each single statistic against the MD label; the label-defining slope and its p-value carry the 0.90.

k input visits	1	2	3	4	5	6	full
PLR2	0.58	0.51	0.68	0.73	0.82	0.82	0.82
PLR3	0.56	0.55	0.74	0.77	0.85	0.86	0.87
MD	0.47	0.47	0.78	0.79	0.87	0.89	0.90

Single statistic	VF slope	slope p-value	last VF level	VF variability	# visits
MD-label AUC	0.91	0.95	0.75	0.71	0.68

cluster-bootstrap CIs) are standard and specified in §3 and §A. The panel is itself rebuilt from the committed per-model analysis outputs (small JSON, versioned alongside the code) by a third script whose reconciliation gate cross-checks every locked constant; only regenerating those outputs from scratch requires the frozen-feature tensors and the raw images on a GPU.

Broader impact. The audit is a safeguard: it is designed to stop an over-optimistic progression-prediction number from reaching a clinical decision. Its risk is the opposite: an over-strict reading could discourage a genuinely useful model, so we frame the checks as necessary conditions for trusting a reported number, not as a ceiling on what is clinically achievable, and we state explicitly that progression *is* predictable at scale in cohorts without these confounds. No patient-identifiable data are used beyond the public release.

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